**Summarization and Simplification of Medical Articles using Natural Language Processing**

**Problem Statement:** The paper addresses the challenge of improving *health literacy* by easing access to complex healthcare information by summarising medical texts and simplifying them lexically by translating specific medical terminology to laymen's terms.

**Algorithms Used:** Used 4 pretrained transformer-based models - *ALBERT, DistilBERT, SciBERT, GPT-2* for extracting summaries from articles and another pretrained model - *en_ner_bionlp13cg_md* for Named Entity Recognition of the medical terms. Used *Lemmatization* for each medical entity and finally *Web Scraping* to generate a simplified meaning.

**Datasets:** Dataset of 400 medical articles taken from the *Cochrane Database of Systematic Reviews (CDSR)*.

**Model Training and Testing:** In generated summary, the medical entities present in the summary are highlighted in green and their respective meaning is displayed to the user when he hovers over the highlighted word. *ROUGE i.e. Recall-Oriented Understudy for Gisting Evaluation* metric has been used to evaluate the summaries produced by the models.

**Results:** The compression rate of the models is 40%. Below figure shows a comparative analysis of the ROUGE scores obtained by the four chosen pretrained models.

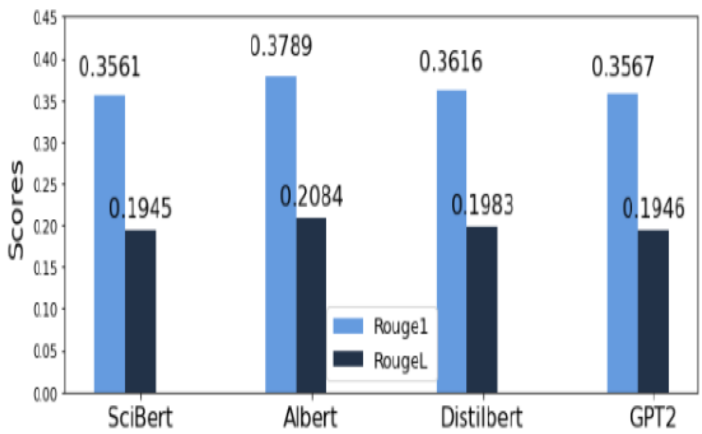


Fig. 5. Comparative analysis of the ROUGE-1 and ROUGE-L scores

**Conclusions:** Authors employ an extractive text summarization approach using ALBERT. Lexical simplification of the summary is done using Named Entity Recognition along with simplifying the complex words through web scraping.

**Open Questions:** The paper suggests future work on improving the paragraph summaries and identification of non-medical complex words and medical abbreviations.

**Relevance to Our Team:** The evaluations into the various transformer based models can guide our team in selecting appropriate models for text summarization which can help clinical researchers to quickly navigate and review documents.

## Extracting Health Evidence Information from Biomedical Literature using LLMs

**Problem Statement:** This article addresses the challenge of bridging the gap between research findings and real-world healthcare applications by exploring *LLMs* to enhance medical practice efficiency and advance *evidence based medicine*.

**Algorithms Used:**
- *INSTRUCTOR* and *OpenAI* for embeddings
- *FAISS* for storing vector embeddings in a knowledge base
- *PICO* Extraction Algorithm(User-Defined)
- *Stuff* for document processing
- *Streamlit* Framework to design an interface for user engagement
- *LangChain* for summarising complex PDFs
- *RAG*-based pipeline
- *RAGAs* for evaluation of retriever and LLMs
- OpenAI LLMs: *GPT-3.5 turbo, GPT-4, GPT-4 turbo, and GPT-4o* to generate responses

**Datasets:**
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3952010/
https://link.springer.com/article/10.1208/s12249-021-02058-y
https://www.thelancet.com/journals/landia/article/PIIS2213-8587(22)00387-4/abstract
https://onlinelibrary.wiley.com/doi/full/10.1002/pst.2020
https://www.sciencedirect.com/science/article/pii/S0753332223014488

**Model Training and Testing:** Overall, 12 PICO queries were generated from the documents. These queries were then passed to each LLM to extract evidence using the PICO framework and answer the queries with the extracted evidence.

**Results:** The average performance of each LLM is detailed in below figure:

| Model | Embeddings | AVG Context Recall | AVG Context Precision | AVG Faithfulness | AVG Answer relevancy |
|---|---|---|---|---|---|
| GPT 3.5 turbo | Instructor | 75.0% | 95.5% | 93.7% | 82.4% |
| GPT 3.5 turbo | OpenAI | 75.0% | 99.7% | 85.00% | 82.3% |
| GPT 4 | Instructor | 85.0% | 95.0% | 96.2% | 85.0% |
| GPT 4 | OpenAI | 75.0% | 100% | 96.7% | 88.3% |
| GPT 4 turbo | Instructor | 71.0% | 94.0% | 95.2% | 87.6% |
| GPT 4 turbo | OpenAI | 90.0% | 99.7% | 89.6% | 87.8% |
| GPT 4o | Instructor | 71.0% | 94.2% | 76.3% | 85.2% |
| GPT 4o | OpenAI | 75.0% | 99.7% | 72.67% | 87.2% |

**Conclusions:**
- LLMs like GPT-4 and GPT-4 Turbo can effectively handle PICO-based queries in systematic medical reviews.
- The choice of embeddings significantly influences performance.

**Open Questions:** The predefined limits on context retrieval can result in incomplete information. Future work should aim to increase the context size to take full advantage of models like GPT-4 Turbo, even though this may come with higher costs which can be addressed by their fine-tuning.

**Relevance to Our Team:** This article is very relevant for our project as it caters to all the aspects of our objective and presents some challenges which we can work on.

**Completing A Systematic Review in Hours instead of Months with Interactive AI Agents**

**Problem Statement:** This article addresses accuracy, relevance and quality issues with existing automatic summarization methods due to lack of domain expertise and intervention. It introduces a human-centered interactive AI agent *InsightAgent* powered by LLMs for SR workflow. This agent also provides visualizations of the corpus and agent trajectories, allowing users to monitor the actions of the agent and provide real-time feedback.

**Algorithms Used:**
- *RSS map* for visualizing corpus structure
- *K-means clustering* for corpus partitioning into semantically distinct clusters
- LLMs: *GPT-4o, Llama 3.3 70B*
- *VA* based interface

**Datasets:** Literature corpus for 15 SRs from PubMed with size ranging from 72 to 7,356 articles

**Model Training and Testing:** Insight agent is evaluated with both the LLMs in autonomous and user interaction modes. Also, comparison is done with existing agents like BM25, ChatCite and AutoSurvey. Evaluation is done by 9 medical experts on the basis of:

- **Record Screening -** Precision, Recall and F1 scores
- **SR Quality** – Rubric based method
- **User Experience –** Feedback Questionnaire and Interviews

**Results:** The average performance of each agent is detailed in below figures:

**4.2.1 Record Screening**

| System | Recall (%) | Precision (%) | F1% |
|---|---|---|---|
| BM25 (Top-100) | 54.30 | 16.50 | 25.3 |
| ChatCite | – | – | – |
| AutoSurvey (Top-100) | 70.30 | 20.43 | 31.6 |
| *Insight*Agent$_{auto}$ (Llama 3.3) | 66.40 | 62.40 | 64.3 |
| *Insight*Agent (Llama 3.3) | 87.90 | **80.10** | 83.8 |
| *Insight*Agent$_{auto}$ (GPT-4o) | 71.10 | 51.90 | 60.0 |
| *Insight*Agent (GPT-4o) | **98.50** | 79.80 | **88.2** |

**Quality of the Generated Summaries**

| System | Score |
|---|---|
| ChatCite (GPT-4) | 47.1 |
| AutoSurvey (GPT-4o) | 54.0 |
| *Insight*Agent$_{auto}$ (Llama 3.3) | 60.9 |
| *Insight*Agent (Llama 3.3) | 70.2 |
| *Insight*Agent$_{auto}$ (GPT-4o) | 62.4 |
| *Insight*Agent (GPT-4o) | **79.7** |

| Question | ***Insight*Agent$_{auto}$ (Llama 3.3)** | ***Insight*Agent (Llama 3.3)** | ***Insight*Agent$_{auto}$ (GPT-4o)** | ***Insight*Agent (GPT-4o)** |
|---|---|---|---|---|
| System was easy to use. | 2.0/5 | 3.9/5 | 2.1/5 | **4.0**/5 |
| Confidence in recommendations. | 2.6/5 | 4.2/5 | 2.8/5 | **4.5**/5 |
| Visualizations aided understanding. | 2.9/5 | **4.2**/5 | 2.9/5 | **4.2**/5 |
| Ability to guide or correct agents. | 2.7/5 | 3.9/5 | 2.9/5 | **4.6**/5 |
| Overall satisfaction. | 3.0/5 | 4.0/5 | 3.2/5 | **4.3**/5 |

**Conclusions:** A single domain expert can finish a high quality SR in only 1.5 hours, rather than months, and reach 79.7% of human-written quality.

**Open Questions:**

- It is a small scale study.
- Considering LLM cost and context length constraint, only title and abstract of each article is used in this study.
- Extraction of numerical statistics or weight evidence is missing.

**Relevance to Our Team:** This article focuses on rapid SRs reducing review time from days/weeks to hours which is one of the main objectives of our project.

**Artificial intelligence to automate the systematic review of scientific literature**

**Problem Statement:** Manually preparing and writing a systematic literature review (SLR) takes considerable time and effort. This paper presents a survey of AI techniques proposed in the last 15 years to help researchers conduct systematic analyses of scientific literature.

**Algorithms Used:**
- *Lingo3G* for document clustering (2013)
- *Fuzzy mining algorithm* to abstract different review models(2018)
- *Voting perceptron algorithm* for supervised learning based paper selection(2006)
- *BoW* representation of the papers(2006)
- *SVM light* for supervised learning based paper selection(2009)
- *TF-IDF (Term Frequency-Inverse Document Frequency)* metric to weight the importance of each word(2019)
- *Logistic Regression* for supervised learning based paper selection(2019)
- *Topic modelling algorithm* (Latent Dirichlet Allocation, LDA) for topic extraction(2019)
- *Ensemble Classifier* for supervised learning based paper selection(2021)
- *Platt scaling LR model* for supporting ensemble(2021)
- *Multinomial NB (MNB)* algorithm to use word count normalisation(2010)
- *Abstrackr* for active learning based paper selection(2012)
- *FASTREAD* for active learning based paper selection(2018)
- *FAST* for active learning based paper selection(2019)
- *DT (ID3 algorithm)* for inferring search strings for paper selection(2017)
- *BioBERT* for encoding sentences in reporting phase of SLR(2020)
- *Long short-term memory (LSTM)* recurrent neural network for transforming encoded sentences into summaries(2010)
- *LBJ* for automatic question generation for report evaluation(2007)

**Datasets:** 34 primary studies are selected from around 9027 references from ACM Library, IEEE Xplore, Scopus, SpringerLink and Web of Science.

**Model Training and Testing:** Not Applicable to this paper as this is a survey.

**Results:** NA

**Conclusions:** Findings reveal a clear interest in applying AI to support paper screening. Regarding other tasks like reporting, the number of studies in these areas is still low.

**Open Questions:**
- One single task i.e. paper selection is predominant.
- AI techniques are still to be explored.
- Specialised algorithms can replace general purpose approaches.
- More active human involvement can benefit AI.
- End-users of SLR automation are not necessarily AI experts.
- AI-based automation of SLRs can be scaled up.
- Performance comparison between different methods and fields

**Relevance to Our Team:** This article focuses on early solutions and classical techniques used for automated SLR which would help us in survey of existing approaches.

## Enhancing Context-Aware Search with Retrieval-Augmented Generation

**Problem Statement:** To provide more accurate and contextually relevant search results from digital content, enhanced retrieval mechanisms are required. This paper explores how SLM-enhanced RAG architectures refine document search results and their implications for future developments.

**Algorithms Used:**
- *BM25* for lexical matching
- all-*MiniLM*-L6-v2 for sentence embeddings
- *FLAN*-T5 for contextual ranking

**Datasets:** The corpus consists of single document, containing unstructured data.
*Document Title: Quantum Computing and its Impact on Cryptography*
*Content: Quantum computing harnesses quantum mechanics to perform computations at unprecedented speeds. One significant impact is on cryptographic security, where quantum algorithms like Shors algorithm can break classical encryption schemes. Post-quantum cryptography is a developing eld aiming to create quantum-resistant encryption methods.*

**Model Training and Testing:** Three retrieval models were evaluated: BM25, SLM, SLM+RAG on the query*: "How does quantum computing affect encryption?"*

**Results:** This experiment measures precision, recall, and response time to determine the most effective retrieval approach.

| Model | Relevance Score | Generated Response |
|---|---|---|
| **BM25 (Lexical Matching)** | 0.500 | N/A |
| **SLM (Semantic Matching)** | 0.7668 | N/A |
| **SLM + RAG (Hybrid Model)** | 0.9202 | "Quantum algorithms like Shor's algorithm can break classical encryption schemes." |

**Conclusions:**
- BM25 is effective for exact term matching and ranking documents based on term frequency and inverse document frequency but lacks semantic understanding.
- SLM Embeddings addresses many of BM25's limitations by employing dense vector representations that encode semantic meaning beyond keyword overlap and hence, improves retrieval accuracy. But it has high computational cost for large scale retrieval and reduced interpretability compared to BM25.
- SLM + RAG integrates retrieval and generative ranking to improve document relevance and contextual awareness, hence, mitigates hallucinations.

**Open Questions:**
- Multi-Document Retrieval for Large-Scale Applications
- Hybrid approaches that integrate BM25 with FAISS and SLM embeddings with FAISS
- Leveraging Domain-Specific Embeddings for Enhanced Retrieval

**Relevance to Our Team:** This article focuses on hybrid model with SLM which can reduce computational cost and latency of our project compared to LLM based frameworks.

**MedBioRAG: Semantic Search and Retrieval-Augmented Generation with Large Language Models for Medical and Biological QA**

**Problem Statement:** The paper addresses accuracy and model hallucination issues with medical Q&A of complex and technical literature. The authors built a system called *MedBioRAG* that searches medical documents, finds the most relevant information, uses fine-tuned model and generates answers based on real medical literature.

**Algorithms Used:**
- *BM25* for lexical matching
- Fine tuned *GPT-4o*

**Datasets:** *NFCorpus, TREC-COVID* for text retrieval testing
*MedQA, PubMedQA, BioASQ* for close-ended QA testing
*LiveQA, MedicationQA, PubMedQA, BioASQ* for long-form QA testing

**Model Training and Testing:** The system was tested on all three types of tasks with respective DBs.

**Results: 1. Text Retrieval -** Semantic search significantly outperformed keyword search.

**2. Close-ended QA -** MedBioRAG achieved higher accuracy than GPT-4o (base), Fine-tuned GPT-4o without retrieval, Previous state-of-the-art models

3. **Long QA** – Responses generated are more coherent and factually correct. Better ROUGE, BLEU, and BERTScore metrics

**Conclusions:**

- Semantic retrieval improves document ranking.
- Adding retrieval reduces hallucinations.
- Fine-tuned GPT-4o + retrieval works better than GPT-4o alone.

**Open Questions:**
- Not validated by medical doctors
- Depends heavily on retrieval quality
- Conflicting documents may reduce accuracy
- Computationally expensive
- Not yet tested on real hospital EHR data

**Relevance to Our Team:** MedBioRAG is structurally very aligned with what we are proposing to build.

Reference: Seonok Kim, "Semantic Search and Retrieval-Augmented Generation with Large Language Models for Medical and Biological QA", arXiv, 2025

https://arxiv.org/abs/2512.10996